Project 2 Slicer

Project Overview: The development of Project2 is focused on using the 3D Slicer software to create a project focused on segmenting and visualizing a volume dataset (from Dicom files). Apart from that, you will have to define some measurements or different markups, that allows to a hypothetic medical expert to get some insights of the volume model. Moreover, you will have to choose a plugin (not explained in the lab sessions) that allows you to work with the volume model chosen by providing advanced tools tailored to specific tasks.

The project will be developed in groups of two people. The project consists of the following tasks:

1. **Segmentation task:** You will choose one of the seven segmentation project options based on the remainder when the last two digits of your DNI are divided by 6. Since you are working as a pair, you have the flexibility to choose between the two options corresponding to the remainders from both of your DNIs. This allows you to decide which project is better suited for your team.
2. **Visualization task:** Using the same volume dataset:
    a. Choosing two different presets for the transfer function, produce different changes into the transfer function to obtain different results.
3. **Measurements task**:
    a. Define a set of measurements or markups that would enable a hypothetical medical expert to gain insights into the volumetric model used in Task 1.
4. **Report task**:
    a. Combine, within a single image, a rendered portion of the volumetric dataset together with the anatomical 2D views, the various measurements or markups generated in the previous task and the models generated from the segmentation task. Create meaningful visualizations that would enable a hypothetical medical expert to gain insights into the key regions of interest within the examined volumetric model. Imagine that these visualizations (snapshots) will be included as part of a report prepared for such an expert.
5. **AI Segmentation task:** Test some of the AI segmentation plugins to segment the same anatomical structures:
    a. Is it possible to use some IA module to obtain the same result? If not, which is the closer approximation that you can obtain automatically? From the obtained approximation, could it be possible to achieve the same result using the Segment

Editor module? Explain how (it is not necessary to compute the segmentation, just only explain the procedure to follow).

6. **Plugin task**: Choose a plugin (not explained in the lab sessions) that allows you to work with the volume model chosen by providing advanced tools tailored to specific tasks.

Project Options

Each option represents a different volume dataset. Calculate your DNI modulus 7 (last two digits of your DNI, divided by 7, and take the remainder) to determine your assigned volume dataset:

- **Option 0:** https://mydisk.cs.upc.edu/s/8TGqiRd6wmWDSYL
  - **Segmentation task:** Segment the intestinal contents: air, homogeneous content, and heterogeneous content.
- **Option 1:** https://mydisk.cs.upc.edu/s/x9yAt2ZG6LebR3y
  - **Segmentation task:** Segment the visceral structures: liver, kidneys, and stomach.
- **Option 2:** https://mydisk.cs.upc.edu/s/GAiWewbX46ejiSE
  - **Segmentation task:** Segment specific bone structures of the skull and jaw, as well as multiple dental elements.
- **Option 3:** https://mydisk.cs.upc.edu/s/ib5CMqeYfEK64Xq
  - **Segmentation task:** Segment the heart and the major blood vessels.
- **Option 4**: https://mydisk.cs.upc.edu/s/xq74Wg2m9mEXwTG
  - **Segmentation task:** Segment the lungs along with the major thoracic bones.
- **Option 5:** https://mydisk.cs.upc.edu/s/7cs7YE9pwTbkiWR
  - **Segmentation task:** Segment the major bones of a single foot.

Delivery instructions:

The project deliverable is due **on January 12th**. The deliverable must be a ZIP file that includes:

- The project files in Slicer that solve the tasks, with meaningful names: Option1_visTask.
- The document that describes how you reach to the solution and answers the questions asked, and the DNI number used for the selection of the volume dataset.
- Please choose the language in which you feel more comfortable to write the document.
- Upload the project to some cloud service (Google Drive, WeTransfer…), and share it with your teacher (eva.monclus@upc.edu). If you share with Google Drive, grant the permissions properly to eva.monclus@upc.edu, there shall be no need of asking permissions for being able to download the file.
- Ensure that everything works when downloaded.
- The ZIP file name must contain the name of the authors. The report also must contain the name of the author.

Use the *atenea* platform to submit your deliverable.